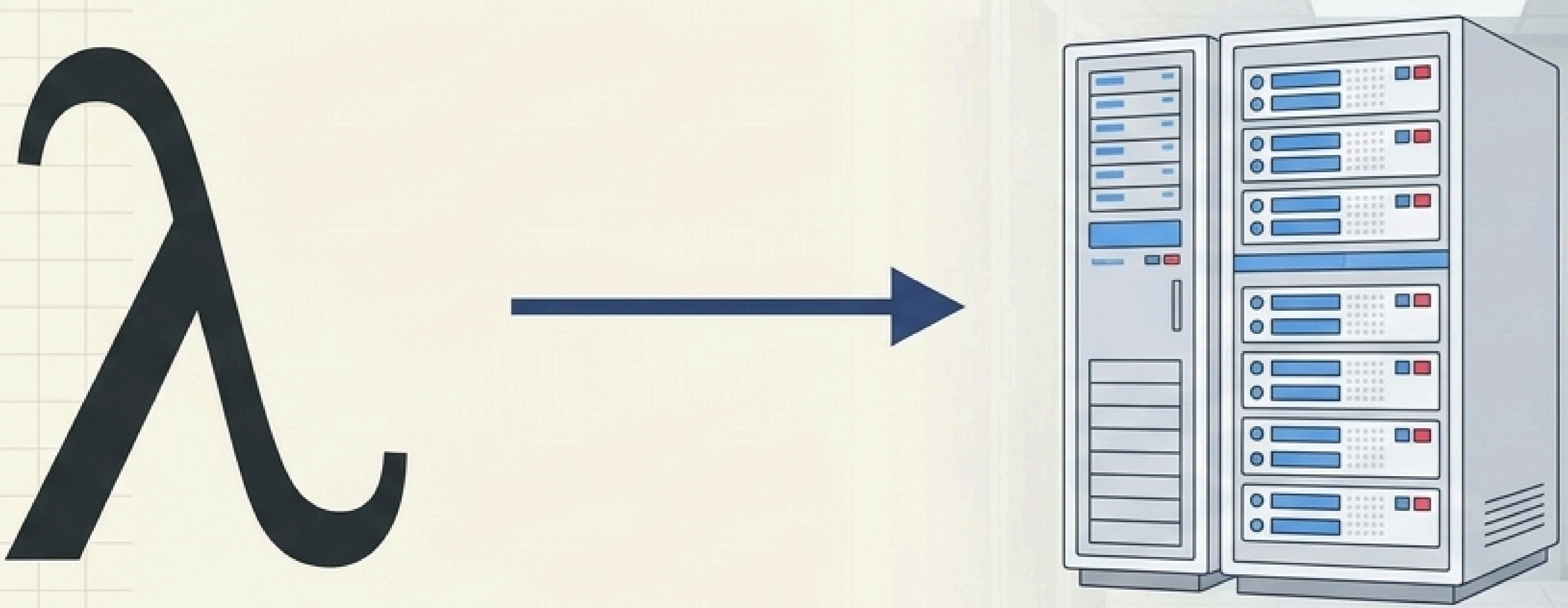
The Power of λ



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Advanced Software Engineering

The Smallest Universal Language

β-reduction

For example:

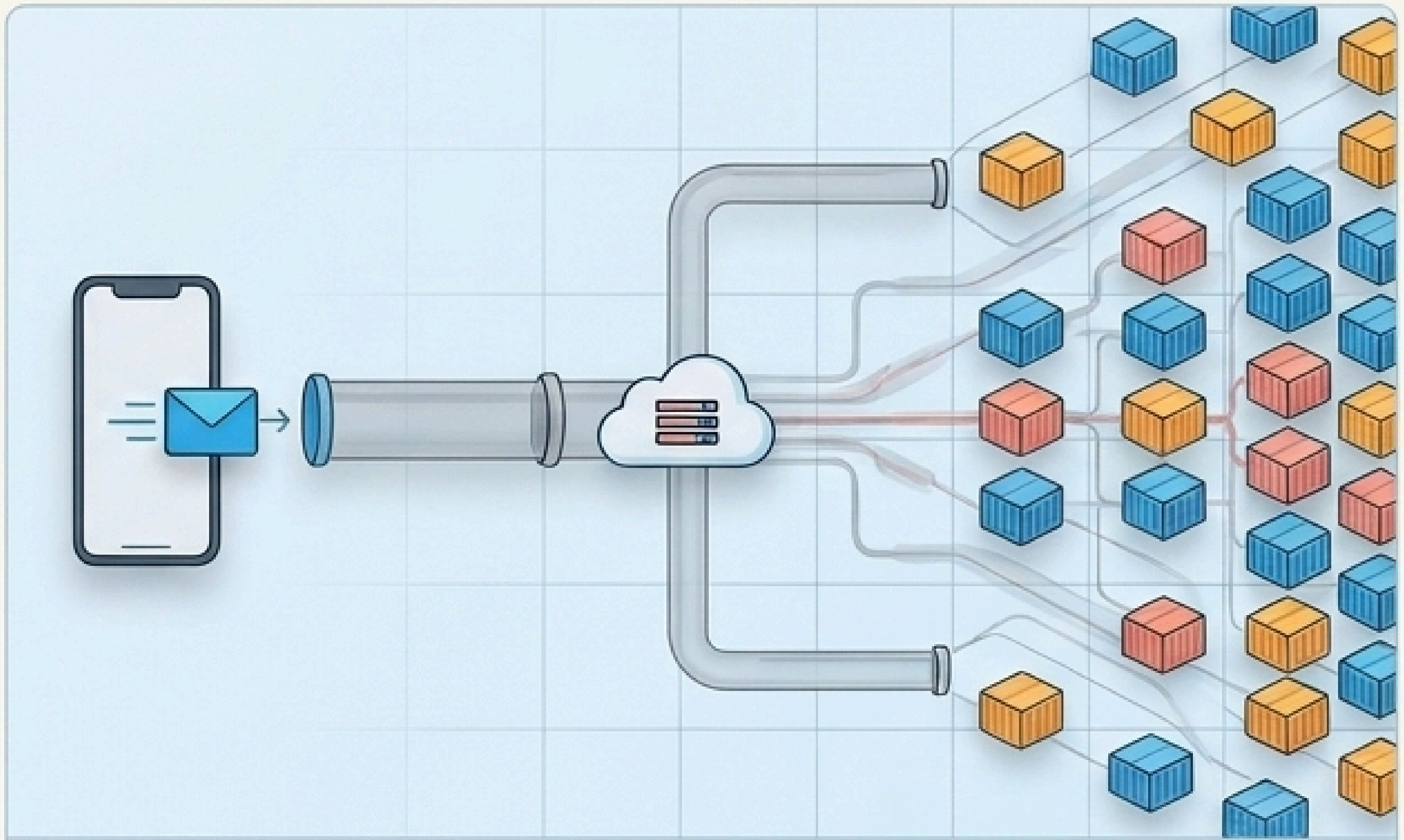
- $(\lambda x.x^2)(5) \to_\beta 25$
- $(\lambda x.y)(z) \quad \to_\beta y$

Definition

- β-redex: A term of the form $(\lambda x.M)N$ (a lamda abstraction applied to another term)
- It reduces to $M[x:=N]$
- The result is called a reduct
- β-reduction is applied recursively until there is no more redexes left to reduce
- A lambda term without any β-redexes is said to be in β-normal form

1936: Computability

1930s Math: Functions as pure, nameless transformation rules.



Today: Infinite Scale

2024 Industry: Stateless, self-contained functions scaling infinitely on demand.

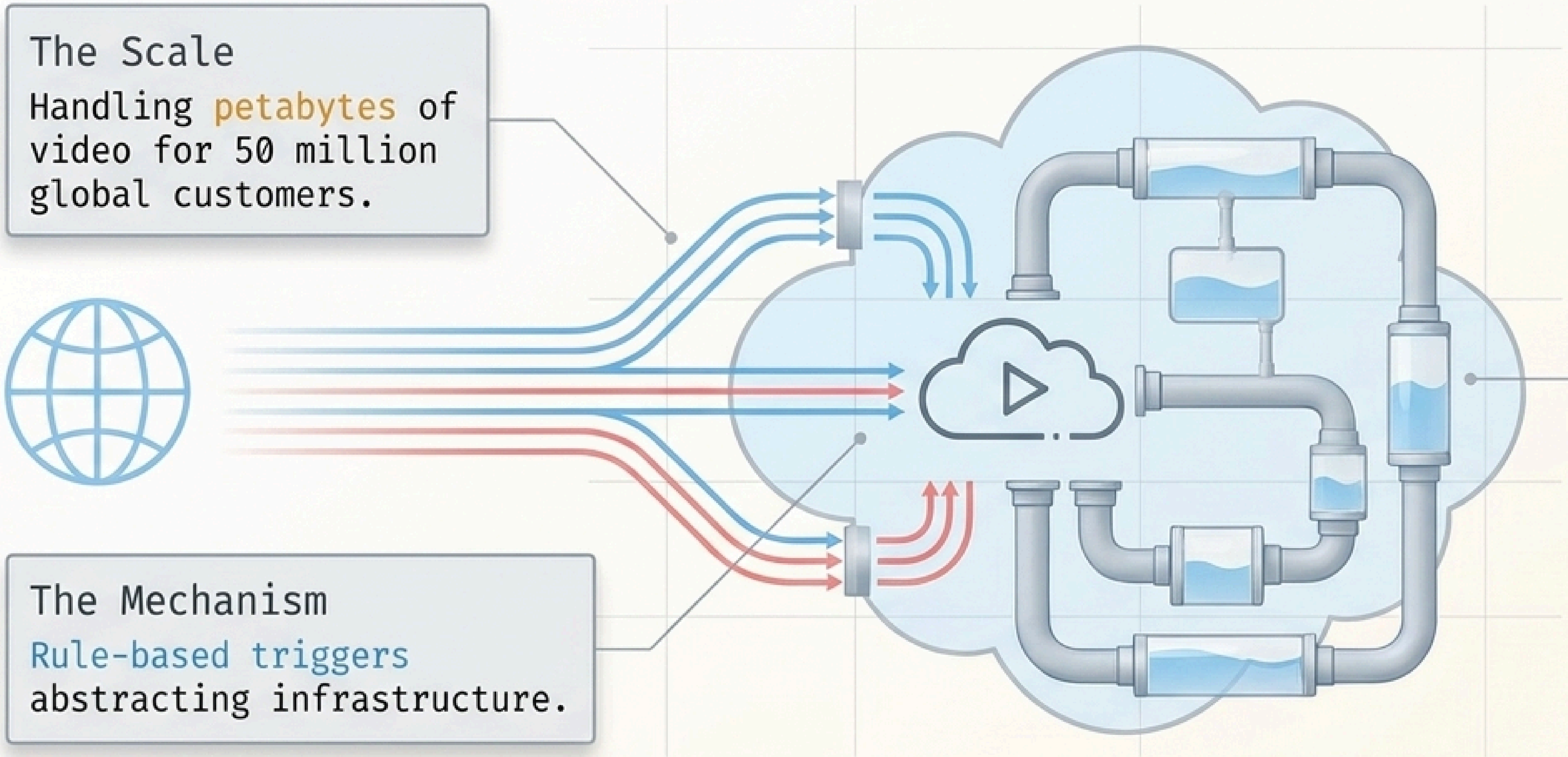
Two Paradigms of Execution

Imperative / Algorithmic	Functional / Declarative
Programmer Focus How to perform tasks step-by-step.	**Programmer Focus** What information is desired.
State Management Mutates data structures in place (High side-effect risk).	**State Management** Non-existent / Immutable (Original data remains untouched).
Primary Flow Control `Loops` and `Conditionals`.	**Primary Flow Control** `Function Composition` and `Recursion`.

Statelessness Enables Global Scale

The Scale

Handling petabytes of video for 50 million global customers.



The Execution

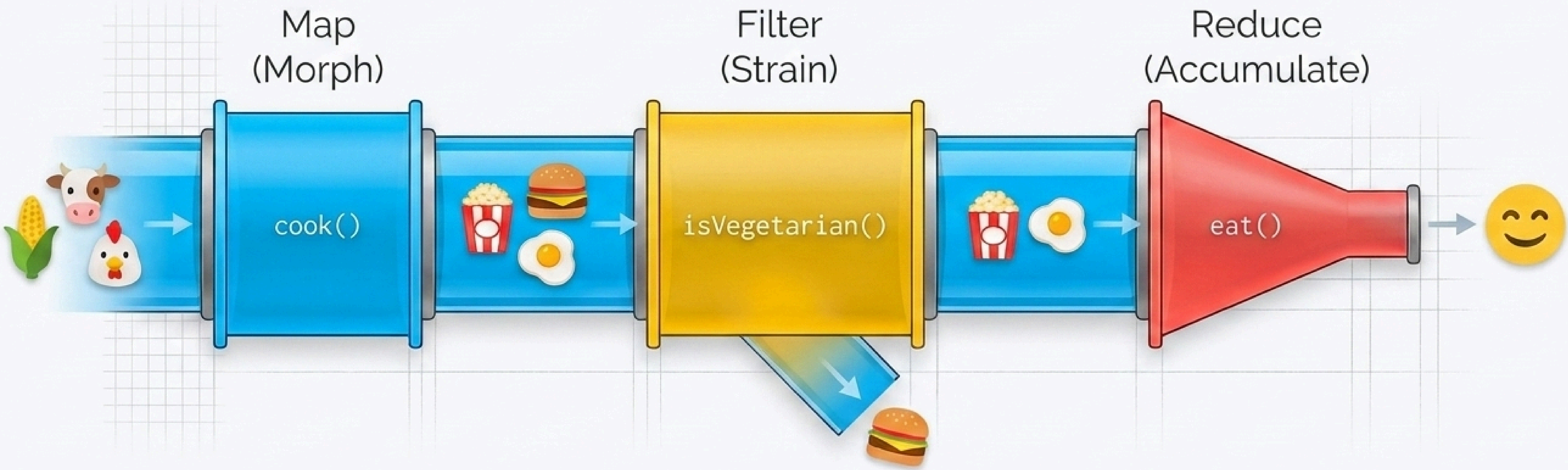
Data written to S3 triggers an isolated Lambda function to evaluate backup routing.

The Mechanism

Rule-based triggers abstracting infrastructure.

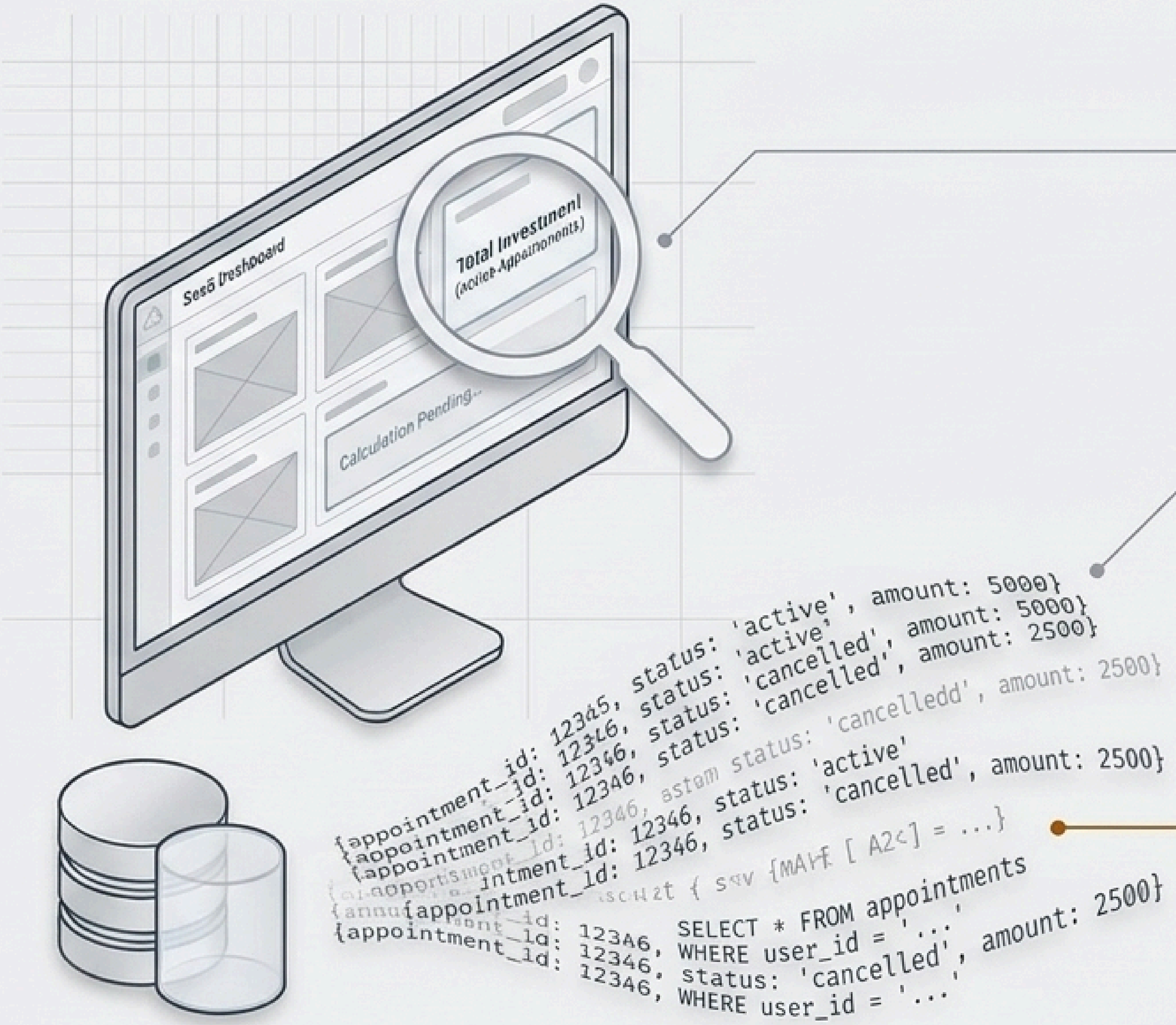
Because Lambda functions are pure and stateless, they require **no shared memory**. They can spin up **millions** of concurrent instances without friction.

The Holy Trinity of Data Pipelines



Notice how the original array is never destroyed. We only flow data through pure functions to create new states.

Real-World Problem: SharkHub Dashboard



Business Need

Provide real-time, zero-latency summary indicators to improve user experience.

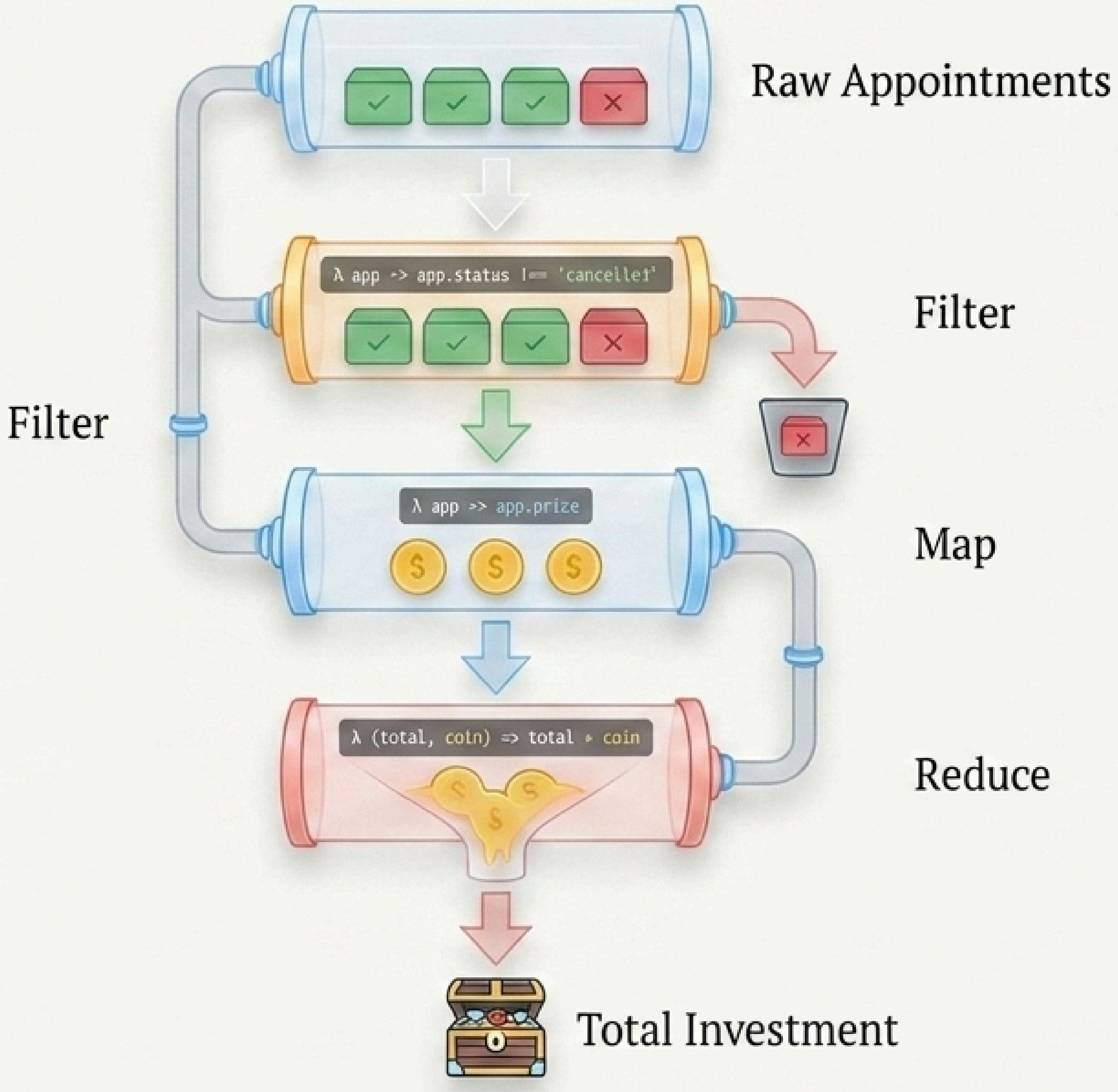
Technical Challenge

The backend supplies a massive array of raw appointment objects (both active and cancelled).

The Imperative Trap

Using traditional loops to parse and calculate this on the front-end risks freezing the UI. We require a highly optimized, functional pipeline.

Designing the Data Flow



Functional TypeScript in Action

Dashboard.tsx

```typescript
const totalInvestment = appointments
    .filter(app => app.status !== 'cancelled')
    .map(app => app.price)
    .reduce((acc, curr) => acc + curr, 0);
```

Immutability
Returns a completely new array; the original appointments array is untouched.

Lambda Expressions
Pure, anonymous functions acting as evaluation criteria.

Composability
Methods are chained together sequentially without temporary variables.

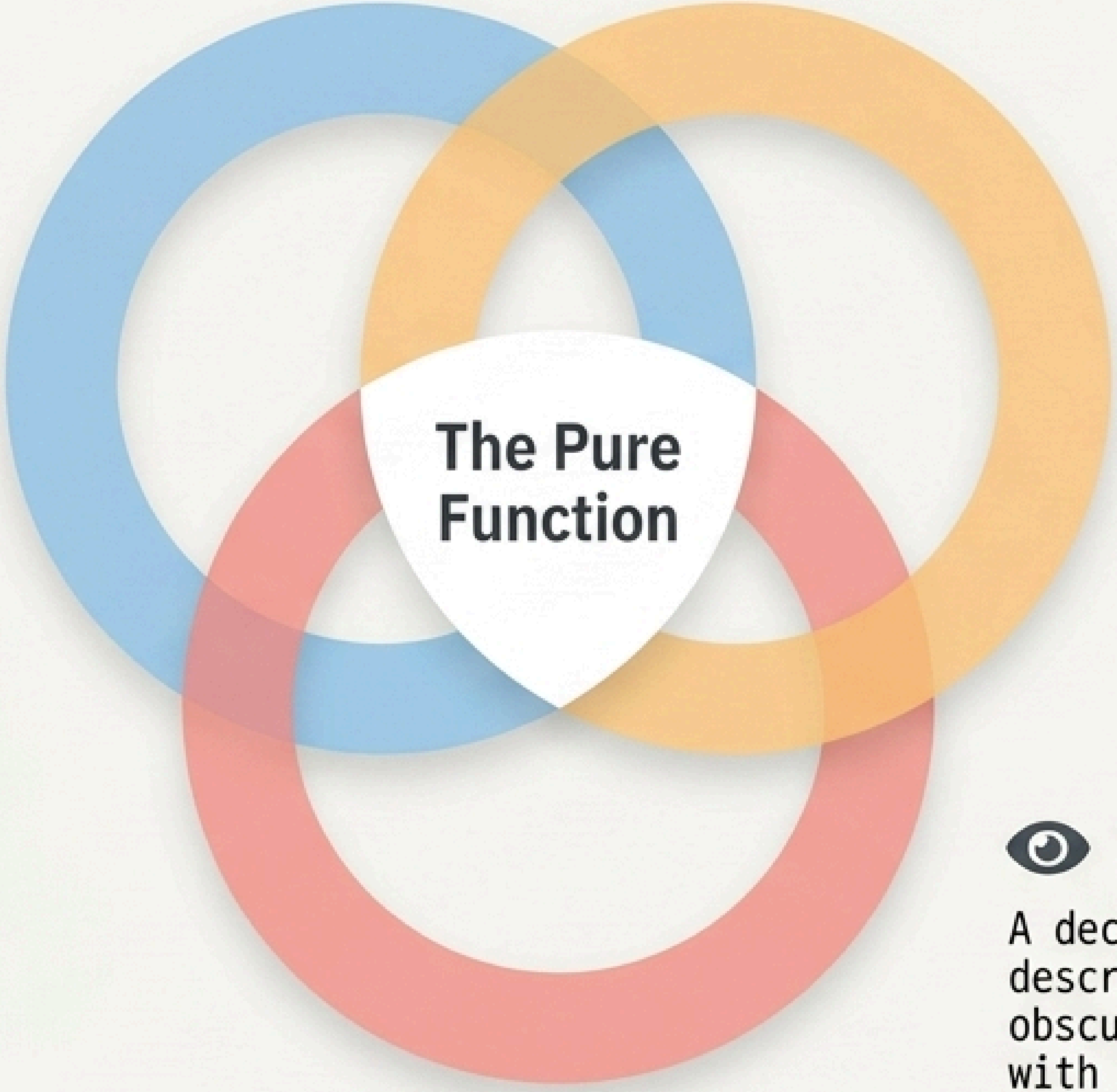
The Functional Advantage

Immutability

Original state is perfectly preserved. Prevents hidden side-effect bugs and guarantees identical inputs yield identical outputs.

Composability

Small, stateless functions can be chained together sequentially, much like structured SQL queries executed directly in the client browser.



Readability

A declarative structure that explicitly describes what to do, rather than obscuring logic behind how to do it with iterative counters.

Thank You

Lambda calculus isn't just theory; it is the blueprint for resilient software.